Canon

PG-50 Black

Sustainable Design Engineering Part I
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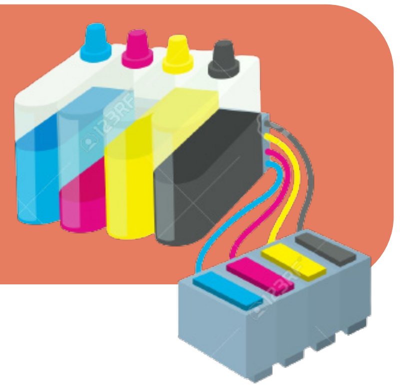
Market Analysis

Key trends and environmental impacts are explored in the initial market analysis, highlighting the need for a shift towards sustainability. Environmental efforts made by both consumers and companies are identified , and key competitors are established.

WHY INK CARTRIDGES?

Printing is an essential aspect of modern life. From personal to proffesional use, printers are used daily. However, the increasing demand for printing, results in an significant amount of waste in the form of ink cartridges. Unfortunately, the majority of these consumables are discarded in landfills, contributing to the global environmental pollution.

However, recycling these items can have benefits for both users and the environment.



MARKET TRENDS

Market Size

The ink cartridge market is a fast growing global market which sells over **65 million** printer cartridges annually in the UK alone. [1]

The printer ink cartridge market size, measured in value, was **$14.10 Bn** in 2022 and is expected to reach **$21.48 Bn** by 2029.

1.1 billion inkjet cartridges are used every year, but **only 15%** are recycled or reused.

New Consumables & Top Trends

-Ink refills

-Recycling and Remanufacturing: brands are now focused on reducing adverse environmental impact.

-CISS Technology: is a cartridge-free printing technology that features refillable ink tanks. Devices equipped with this technology contain two years of ink in the tank

-Alternatives For Ink and Toner: They're developing new materials to use as alternative for ink and toner in order to make more eco-friendly products.

-Subscription/Contractual-Based Supplies Replenishment Programs: useful for people who are not technology friendly which reduce cost but also allow users to use the printer sustainably. [3]

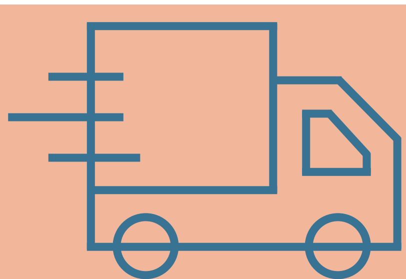
Consumer Cartridge Disposal

Consumers often don't know where to throw away their empty cartridges, which leads to waste stream contamination, poor quality of any recycled material and **350 million** consumables ending up in landfills each year. [1]

3 pints of oil are used in the manufacture of a new cartridge and this dangerous plastics and polymers can take up to **1000 years** to decompose.[1]

Sustainability Schemes

Nearly all printer cartridge manufacturers offer free recycling schemes for their consumables as a courtesy to their consumers. Alternatively, other current recycling schemes like InkRecycling [4], run a free postal service and reimburse users for their empty cartridges. Additionally, alternate markets like therecyclingfactory, offer recycled cartridges for up to **70% less** than the original brand cartridges[1]



KEY COMPETITORS

EPSON hp RICOH imagine. change. Canon brother at your side

The printing insdutry is a very competitive market, with North America and Europe being the leading regions. Some of the key players include: HP Development Company, Seiko Epson Corporation, Ricoh, Brother Industries and Canon Inc. [2]

They each offer their own inkjet cartridges usually designed for their own printing machines, and they're continuously improving their cartridges ink quality, price and sustainable practices.

This project will focus on Canon's higher-end ink cartridge consumables, as they are a worldwide company with stable manufacturing processes and active recycling schemes, who have clear future goals but haven't yet reached them.

KEY INSIGHTS

Compatibility

Many brands printers are only compatible with their own genuine brand consumables. So even if users wanted to purchase more eco-friendly products or subscribe to more sustainable schemes, it might prove very challenging and discouraging.

Difficult Product Disposal

Many users are discouraged by the cartridges difficult separation for recycling.

Eco-friendly Products

Cartridges that can be recycled and remanufactured have been in high demand in 2022.

Market Comparison

The brand in question, Canon, is explored in more detail, showing their recent sustainability efforts and future goals. Shortcomings in the top cartridges brands allows Product's sustainability comparison.

Canon

History
Canon is a leading technology company founded in Japan in 1937. Some of their future goals include becoming a carbon net zero company by 2050, to do so they're improving their designs , keeping products and materials in use for longer, making consumables 100% recyclable/reusable and eliminating single-use plastic and polystyrene from packaging. [5]

Product Quality
Canon advertises their genuine consumables by focusing in 4 main aspects: [6]

-35% more high quality prints than imitation cartridges
-100% reliable, print with less stress
-100% consistent quality prints
-Machine good as new when using genuine brand

LONGEVITY

DURABILITY

UNRIVALED QUALITY

KEEP IT CANON

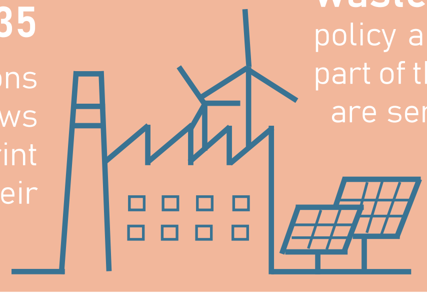
Sustainability
Canon assumes responsability for the recycling and recovery of raw materials from their cartridges and has been collecting and recycling used ink cartridges since **1996.** [7]

2008-2021

•**42413 tons** recycled as raw materials

•**33619 tons** reused directly.

Recycling Schemes
Canon offers a free recycling programs operational in **35 countries** and regions worldwideas of 2021, which allows users to download and print Freepost labels to return their consumables for recycling. [5]



Committed to a **'no waste to landfill'** policy and assures no part of their cartridges are sent to landfills.

A LOOK AT THE MARKET

	Canon	EPSON	Kodak	hp
Price	£ 19,63	£ 20,78	£ 10	£21,64
Recycling Schemes	Free recycling program allows users to download and print Freepost labels for to return their consumables for recycling and reuse	Established collection and recycling programs consisting of single return (via postal) and bulk return (via box collection)	Doesn't run a recycled service. However, they offer a refill kit, which allows you to replace the ink in your existing cartridges	Runs a free recycling system with Planet Partners offering postage-paid envelopes or labels to return HP cartridges by post or allocates boxes for compamies to return their cartridges in large quatities
Product				

INSIGHTS

Quality is a Major Factor in Purchase
Most consumers highly value quality when purchasing ink cartridges. They are wary of recycled plastic and reused components, so the high quality of product should be maintained.

Combine Delivery & Collection Schemes
Not all brands offer non-time consuming recycling programs for individual users.
When cartridges are delivered to consumers, there is an excellent opportunity to return the empty cartridges.The delivery drivers can easily collect and return them to the manufacturer.

Incentivise Recycling
Although eco-friendly cartridges are on high demand. Additional incentives would be very beneficial, as consumers respond well to financial incentives. So companies should focus on rewarding eco-friendly users with discounts and reducing the price of recycled cartridges.

The Users

Two user groups of different sustainability effort are represented below. Their needs, pain points and motivations are important to consider by brands who wish to satisfy their consumers while achieving their sustainable goals

BIOGRAPHY

Age	35
Occupation	Personal assistant
Employment	Full time
Location	Manchester

NEEDS

-Keep printer working at all times
-High quality ink
-Minimising waste

PAIN POINTS

-Refilling Challenges
-High cost & commitment for reusable products
-Compatibility issues

MOTIVATIONS

Sustainability	●●●●
Price	●●●
Convenience	●●●●
Quality	●●●●●

BIOGRAPHY

Age	21
Occupation	University student
Employment	Full time
Location	London

NEEDS

-Time efficiency and convenience
-Keep spending to a minimum
-Shipping and delivery options

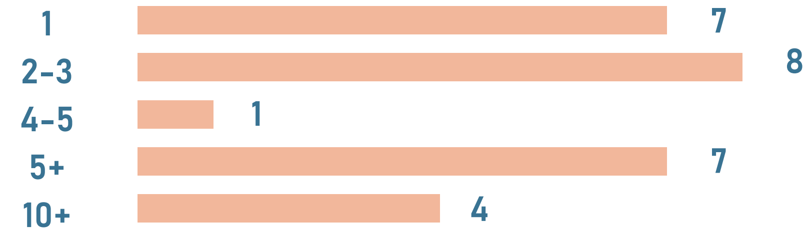
PAIN POINTS

-Printer compatibility
-Feeling forced to use unsustainable cartridges due to their convenience
-High price for better quality

MOTIVATIONS

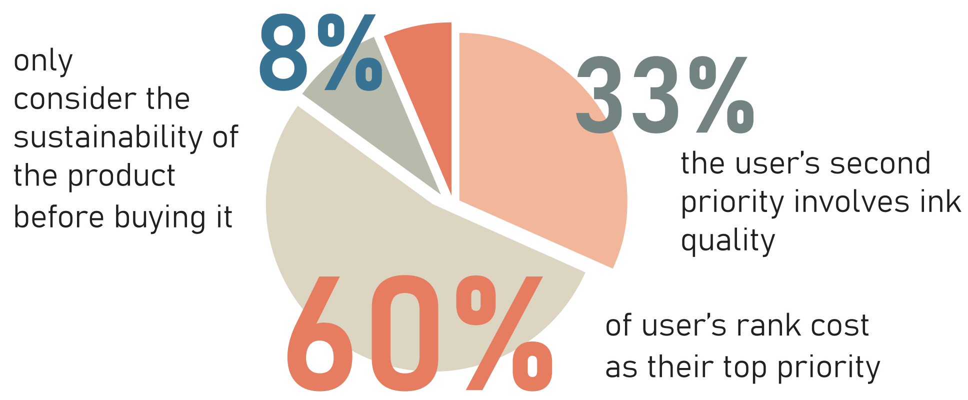
Sustainability	●●
Price	●●●●●
Convenience	●●●●
Quality	●●●

How many ink cartridges do you use per year?

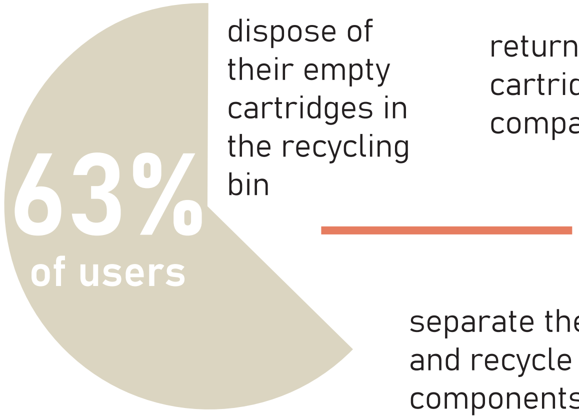


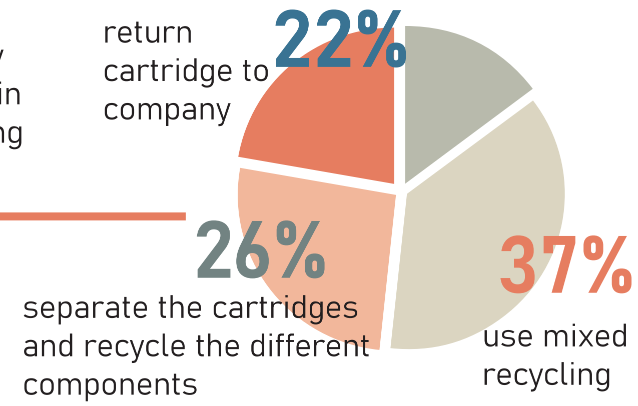
There is a very notable difference in ink cartridges purchases between different age groups. Younger users tend to buy between 1-3 ink cartridges, whilst older users buy between 5-10

What do you look for when buying an ink cartridge?



How do you dispose of your empty cartridges?





INSIGHTS FROM USER QUESTIONERS

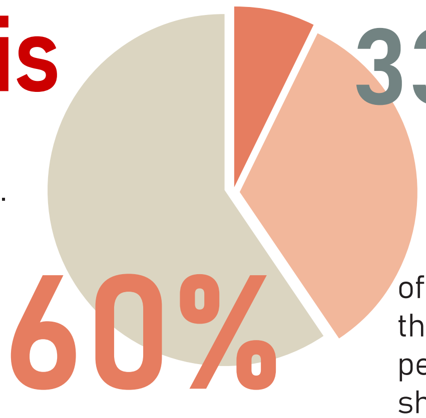
- The more stable income and settled users consume double the amount of products, so the cartridges should be tailored to their needs, whilst also satisfying younger users needs.
- Affordable price and high quality are key features for users when selecting their cartridges.
- More than half users recycle their cartridges but most don't take the time to recycle the components separately, so easy disassembly is key.
- Brand recycling programs are not commonly known and should be more advertised to inform users about all their sustainable options.

User Journey and Analysis

The majority of the UK purchasers of ink cartridges do so in person , at least 4 times per year, so the persona analysis focused on offline buyers.

User interviews and survey results revealed 3 key user priorities:

Cost | Time | Quality



33% of users buy their ink cartridges from online retailers

60% of users prefer to buy their ink cartridges in person at their local shop

An increase in sales through online retailers can be noted, so brands should update and expand their virtual presence to match their sustainable conscience

Brands should refocus on visually sustainable packaging and in-box recycling options as users rely on visual displays during their purchase

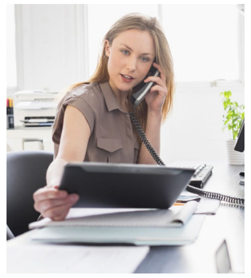
The user's journey in using the Canon black cartridge is displayed below. Their trends, eco-awareness and feelings are mapped throughout, which are used to derive new opportunities.

PERSONAL ASSISTANT

'Overworked, busy, diligent'

-Full time assistant
-Relies on printer to convey information

Printing trends:
-has a printer in office
-print multiple daily documents
-buys cartridges every month



Eco-aware
"Our company has a firm belief on recycling and minimising our environmental impact"

UNIVERSITY STUDENT

'Un-organised, stressed,social'

-Centres printing depending on work-load
-Relies on shared printer

Printing trends:
-Uses printer especially during exam season
-Shares printer
-Price is a priority

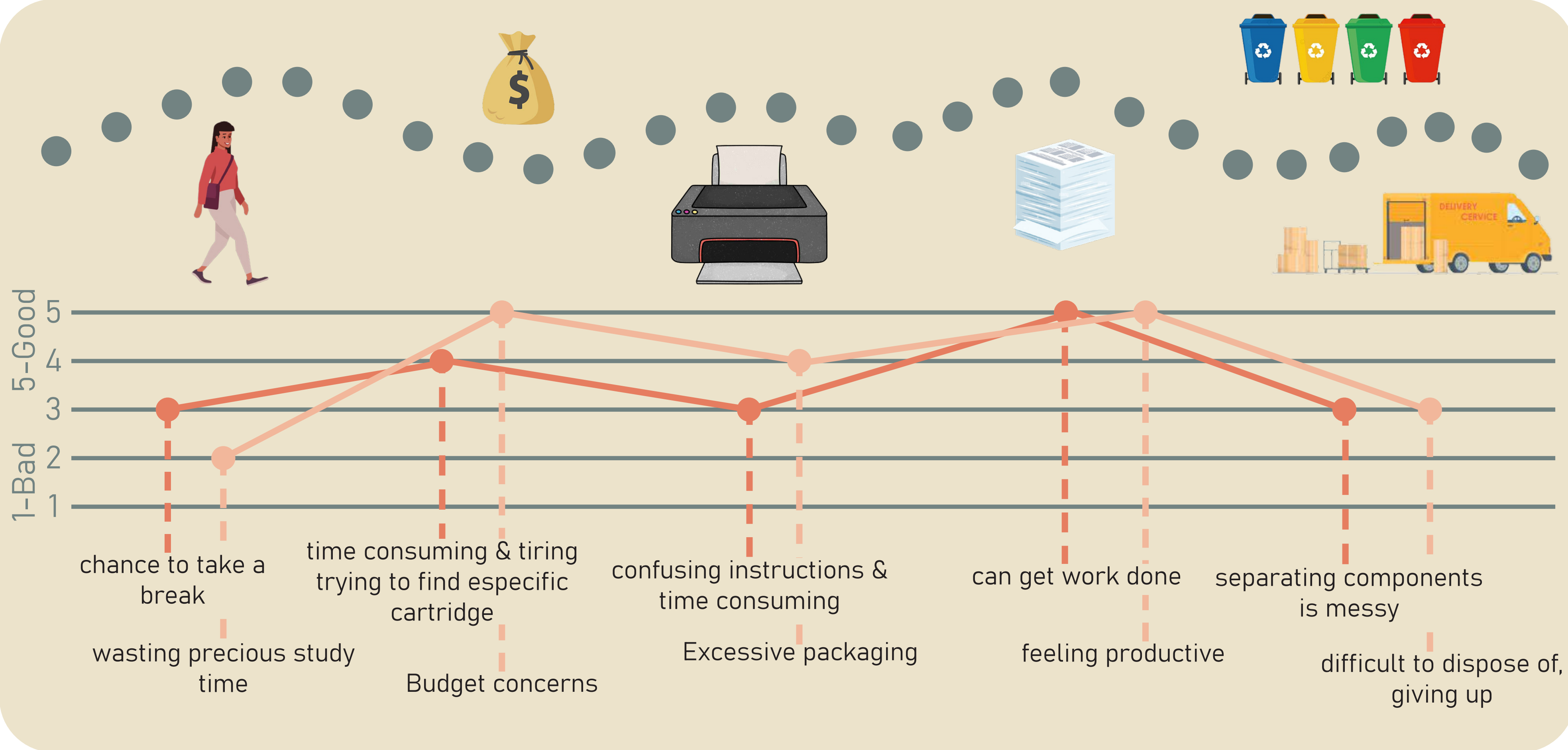
Eco-passive
"I barely have time to buys new cartridges, much less recycle them"

TRAVEL | PURCHASE | REPLACING | USE | DISPOSAL

ACTIONS

EXPERIENCE

THOUGHTS



OPPORTUNITIES

Printing shops are less common so users must dedicate time and resources to get product, so visual incentives or house delivery options are crucial

Affordability rises and the product's sustainability falls on the users priority list, so cheaper options are essential.

Users don't have the space or desire to keep wasteful packaging used to beautify product, so small boxes and minimising the amount of materials is key

Greatest user satisfaction with product due to high value results, so maintaining quality should be a priority

Users want to easily and quickly dispose of used cartridges, so easy separation for recycling or access to home retrieval are critical

Product Teardown

To look at components and materials of the product in more detail, a teardown was carried out. Disassembly and main structural and informational features are analysed. The design for disassembly and recycling guidelines are used to check the recyclability of the product

1+2 Background layer

Material: Paper
Purpose: Promote brand and provide proffesional back-ground for finial product
Mass: 2.5g

3+4 Brand history & instructions

Material: Paper
Purpose: Informs user on how to insert product
Mass: 10g

5 Protection case

Material: Polypropylene - **injection molding**
Purpose: Protect and cushion cartridge case
Mass: 12g

6 Protection case lid

Material: PS
Purpose: Seal protection case to prevent damage
Mass: 0.2g

7 Cartridge lid

Material: PPE+PS - **injection molding**
Purpose: Seal cartridge case to prevent ink spillage
Mass: 6g

8 Protective tape

Material: PE
Purpose: Tape cartridge to prevent ink from leaking
Mass: 0.1g

9 Cartridge case

Material: PPE+PS - **injection molding**
Purpose: Store ink sponge
Mass: 20g

10+11 Sponge reservoir

Material: Hydrophobic foam
Purpose: Hold ink and repel outside water or humidity in the air
Mass: 4g

12 Electrical contacts

Material: Gold-plated copper
Purpose: Connect to printer to enable communication and monitor ink level
Mass: 0.2g

13 Plastic case

Material: PS
Purpose: Extra protection
Mass: 20g

14 Main box

Material: Cardboard
Purpose: Protect and enclose product. Display brand
Mass: 16g

DESIGN FOR DISSASSEMBLY?

- Added glue used to seal cartridge case creates a strong adhesion, difficult to separate without proper tools ✗
- Tightly packed sponge made separation harder and messier due to ink leftovers ✗
- Small crevices, irregular insides due to injection molding and glue leftovers made complete sponge removal very hard ✗

DESIGN FOR RECYCLING?

- The main cartdige case components and electrical contacts can be adequately recycled in the Canon facilities ✓
- The foam will probably end up in a landfill as it can't be disposed of in the recycling bin ✗
- The plastic case and protection case lid can't be recycled and should be placed in the waste bin ✗
- The paper components are easily recycled ✓

ISSUES & INSIGHTS

- After initial inspection, packaging for this product is not the main concern as it can be easily replaced with a more sustainlable material.
- No information symbols where provided in the packaging and therefore no disposal instructions where available.
- Cartridge itself is difficult to disassemble for further recycling
- **Fatal flaw**: many of the plastic materials are mixed and are therefore complicated to recycle

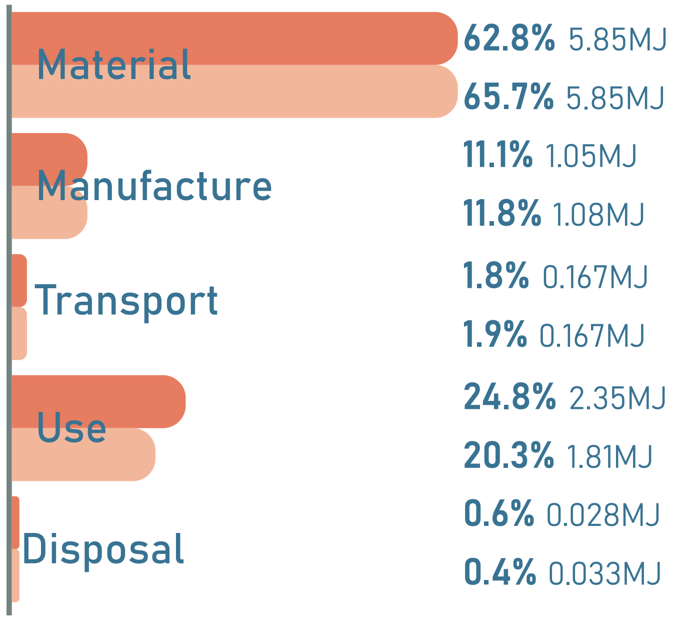
Life Cycle Assesment

Component	Material	Recycled content* (%)	Part mass (kg)	Qty.	Total mass (kg)	Energy (MJ)	%
Background layer	Paper and cardboard	Virgin (0%)	0.0025	1	0.0025	0.071	1.2
Instructions	Paper and cardboard	Virgin (0%)	0.01	1	0.01	0.28	4.8
Protection case	Polypropylene (PP)	Virgin (0%)	0.012	1	0.012	0.81	13.9
Protection case lid	Polyethylene (PE)	Virgin (0%)	0.0002	1	0.0002	0.016	0.3
Cartridge lid	Polystyrene (PS)	Virgin (0%)	0.006	1	0.006	0.49	8.4
Protective tape	Polyethylene (PE)	Virgin (0%)	0.0001	1	0.0001	0.008	0.1
cartridge Case	Polystyrene (PS)	Virgin (0%)	0.02	1	0.02	1.6	27.9
Sponge reservoir	Rigid Polymer Foam (LD)	Virgin (0%)	0.004	1	0.004	0.32	5.5
Elctrical contacts	Copper alloys	Virgin (0%)	0.002	1	0.002	0.13	2.2
Plastic case	Polystyrene (PS)	Virgin (0%)	0.02	1	0.02	1.6	27.9
Main box	Paper and cardboard	Virgin (0%)	0.016	1	0.016	0.45	7.7
Total				**11**	**0.093**	**5.9**	**100**

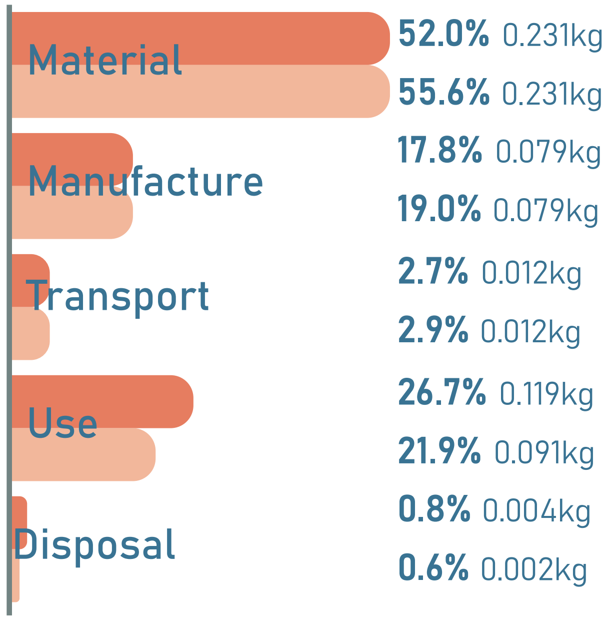
[8]

recycled product   not recycled product

ENERGY ANALYSIS



CO2 FOOTPRINT ANALYSIS



END OF LIFE POTENTIAL

-2.2MJ	-0.0789kg
-0.175MJ	-0.00029kg

Recycled and not recycled products where analysed and a significant increase in the energy and CO2 potential is visible when recycling the product.
As 96% of the recycled cartridge material is recovered back into the product, these values are very acurate.

INSIGHTS

The product has relatively low embodied energy and CO2 footprint for its materials and manufacturing processes. Assuming that the cartridge is sent to a Canon facility to be recycled, the consumable will have a nearly perfect circular loop. If it is sent to TerraCycle and is 100% recycled, the end of life potential also seems promising. However, the fact that the plastic packaging is difficult to recycle and will most propably end up in landfills is a major problem. Additionally, although the materials that can't be recycled are used as energy recovery, ensuring no waste, some alternative recyclable materials should be explored.

Further analysis using the Eco Audit toll is carried out and analysed below. After establishing that the plastic and cardboard packaging can be easily reduced and recylced,a material circularity is conducted to pinpoint its and the cartridge's cycles.

ASSUMPTIONS

Recycled content:
·Original, unrecycled ink cartridge

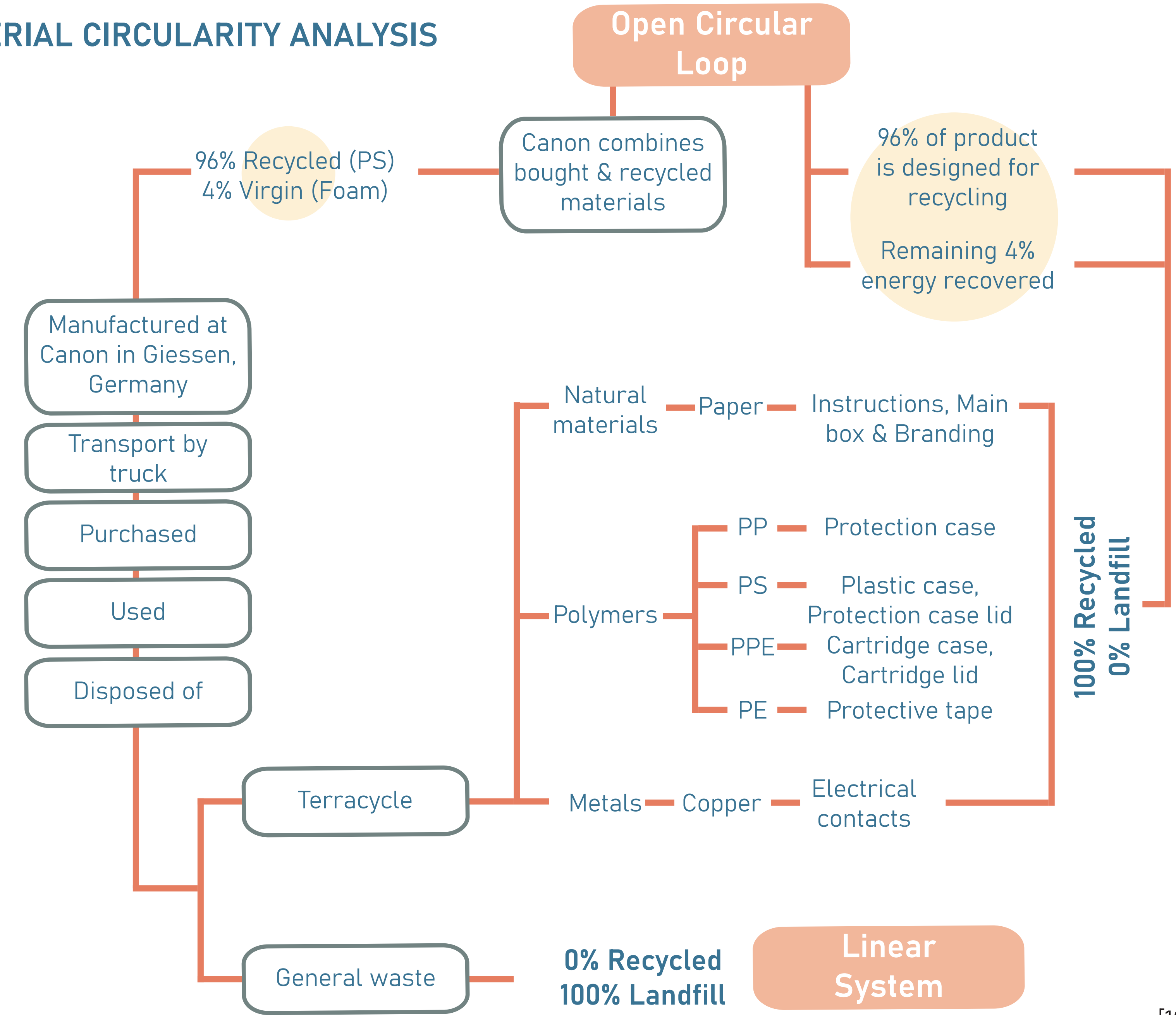
Recycled at Canon:
·Cartridge Body and cartridge lid

Life time:
·1 month

Transport:
·After being recycled in the Canon facility in Giessen, Germany, [9] it was assumed that the cartridges where transported to be manufactured at the Canon Production Printing at Poing, Germany. They are then distributed to Ryman at High Street Kensington, London.

– 458 km Road Freight (Giessen to Poing, Germnay)
–744 km Road Freight in 14 tonne (2axle) truck (Poing, Germany to London, UK)

MATERIAL CIRCULARITY ANALYSIS



[10]

Flowmapper Configurator Board

The cartridge's system, which is the only component that involves a circular loop, is explored in more detail using a flowmapper.

RESOURCE

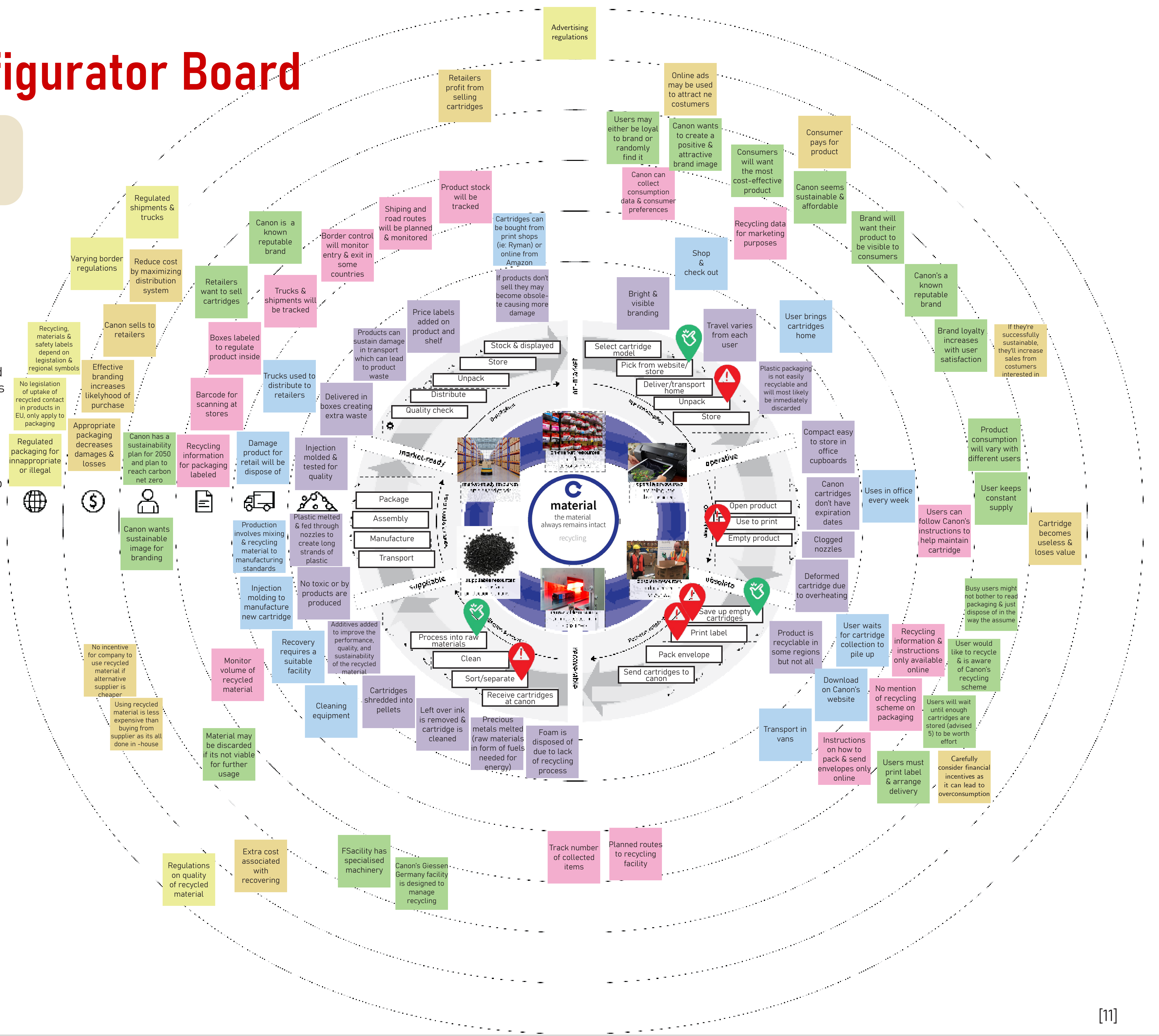
Product	Component	Material
Canon PG-50 Black cartridge	All elements are considered in the flowmapper but the focal point is the ink cartridge case	The cartridge body is made from various plastics (PPE+PS) and filled withother materials

CONSUMER

Needs	Function	Who are they?
Maintain high quality printing & feel that they're reducing their environmental impact	Supply ink to create text, images or graphics on printing media	People over 18 who want to print high quality work

CONTEXT

Geography	Consumption	Intensity
The product is distributed internationally but this analysis will be EU focused	Consumption takes place at the office, either at home or the work place	Cartridges usually last about 1-3 months when used regurlaly



[11]

Pivotal Analysis

The most pivotal points on the flowmapper are selected and jusified

Process	Scope of process	Why is this process pivotal?	Relevant elements						
Pick from website/store	The product is picked & bought by the user	Canon is a high-end brand, with a working recycling scheme compared to other brands which increases the chance of users re-purchasing and increasing brand loyalty	Bright & visible branding	Shop & check out	Recycling data for marketing purposes	Consumers will want the most cost-effective product	Canon seems sustainable & affordable	Brand will want their product to be visible to consumers	Canon's a known reputable brand
Unpack	Plastic & Cardboard packaging will be opened & discarded	The product lacks information on how to dispose of the packaging. Since the type of plastic isn't specified, there's a higher chance of incorrect disposal & waste stream contamination	Plastic packaging is not easily recyclable and will most likely be immediately discarded	Brand loyalty increases with user satisfaction	If they're successfully sustainable, they'll increase sales from costumers interested in sustainability				
Empty product	Cartridge no longer works	Consumable will stop working due to various reasons, but Canon's recycling scheme will only accept empty cartridges, so damaged or defective cartridges won't be recycled	Canon cartridges don't have expiration dates	Clogged nozzles	Deformed cartridge due to overheating	Users can follow Canon's instructions to help maintain cartridge	User keeps constant supply	Busy users might not bother to read packaging & just dispose of in the way the assume	Cartridge becomes useless & loses value
Save up empty cartridges	The user collects used cartridges to send to Canon	The user is aware and is taking part of Canon's recycling scheme, rather than tossing the used razors in the general waste with the packaging. However, this is not marketed efficiently, which can turn into a missed opportunity	Product is recyclable in some regions but not all	User waits for cartridge collection to pile up	Recycling information & instructions only available online	User would like to recycle & is aware of Canon's recycling scheme	Users will wait until enough cartridges are stored (advised 5) to be worth effort		
Print label	User goes onto Canon's website to download & print shipping label	Although this system encourages sustainability as users can access the label from the comfort of their home, some might feel discouraged if they run out of ink & have to wait until the next shippment arrives	Download at Canon's website	No mention of recycling scheme on packaging	Instructions on how to pack & send envelopes only online	Users must print label & arrange delivery	Carefully consider financial incentives as it can lead to overconsumption		
Sort/separate	Canon disassembles & separates plastic components	Eventhough 96% of the product is recycled, there are still parts which aren't recyclable and will pollute the environment unless they end in different recycling schemes	Foam is disposed of due to lack of recycling process	Precious metals melted (raw materials in form of fuels needed for energy)	Facility has specialised machinery	Canon's Giessen German facility is designed to manage recycling			
Process into raw materials	Plastics are shredden into pellets	The recycled materials can gain back value and returned back to the product, which decreases brand's need for virgin materials	Cartridges shredded into pellets	Additives added to improve the performance, quality & sustainability of the recycled material	Recovery requires suitable facility	Monitor volume of recycled material	Material may be discarded if its not viable for further usage	Extra cost associated with recovering materails	Regulations on quality of recycled material

Insights & Opportunities

Technical, economic and environmental challenges and their opportunities are discussed, referencing previous data and insights.

What is working well?	What isn't working?	INSIGHTS & OPPORTUNITIES
Canon's combination of being a high-end brand whilst having an accessible recycling scheme provides a circular, closed loop, with no wasted materials or energy, which is very attractive to users and ensure brand loyalty.	Lack of material especification and disposal instructions in packaging hinders recycling process and increases incorrect waste disposal and waste stream contamination.	All consumables should i**nclude material and recycling information** to encourage users to pursue a more sustainable lifestyle
User reasearch highlighted the importance of price and quality for most consumers, which Canon achieves without fail.	Canon's recycling scheme is only available online and not advertised enough, so many users don't know about its existence and can't therefore participate, even if they want to be more sustainable. Additionally, Canon has no sustainable advertisement on packaging, which decreases its desirability for new users	**Advertisement in the packaging** is essential to attract new users, maintain brand loyalty and keep up to date with new trends. Canon should promote its green inititives not only online but through their consumables. The brand's **recycling scheme** should also be **added** to the packaging without adding unnecessary materials.
Easy recycling scheme system encourages users to become more sustainable without having to give up their time or resources.	Packaging of product is excessive and difficult to recycle as it has too many protective measures and the plastics used are mixed and non-recyclable. Additionally, the packaging is very bulky and consumers have no desire to carry or store unnecessary materials.	Packaging should be made from an **easily recyclable material** and have **minimal layers** to ensure minimal waste creation.
	Cartridges that are not empty won't be accepted into Canon's recycling scheme, which means they'll end up in the landfill if the user can't rely on a third party company, with a different recycling program, to achieve their sustainability goals.	The Canon brand should **account for damaged consumables** and i**mplement additional steps in their recycling scheme** to accomodate for those users and decrease their brands contribution to environmental pollution.
	Some of the cartridges materials, like the foam, aren't recyclable and will most likely end up in a landfill unless they're recovered by other recycling schemes	The brand should **move away from un-recyclable plastics and foams** to ensure the recovery of the material even if it doesn't end up in their brand's recycling scheme.
	Cartridges are difficult and messy to separate to recycle into its different components if consumers use customary recycling bins	Canon should **simplify the cartridge's disassembly** without compromising the products quality to facilitate the product's dismantling. This could be achieved by altering the product's design

References

[1] therecylcingfactory.com-Cartridge Facts, Reasons why whe recycle cartridges [online] Available at: https://www.therecyclingfactory.com/facts

[2] maximizemarketresearch.com - Printer Ink Cartridge Market, Global Industry Analysis and Forecast (2023-2029) [online] Available at: https://www.maximizemarketresearch.com/market-report/global-printer-ink-cartridge-market/68567/#:~:text=Who%20are%20the%20key%20players,%2C%20Ltd.%2C%20Canon%20Inc.

[3] (January 17)Inkhousedirect.com - Recent Posts, Printer ink Cartridges Market Latest Trends 2022 [online] Available at: https://www.inkhousedirect.com.au/printer-ink-cartridges-market-latest-trends-2022/#:~:text=Continuous%20Ink%20Supply%20System%2C%20commonly,of%20ink%20in%20the%20tank.

[4] Inkrecycling.org-Ink Cartridge Recycling & Our Environment [online] Available at: https://www.inkrecycling.org/environment/

[5] www.canon.co.uk - Sustainability at Canon, Reducing environmental impact [online] Available at: https://www.canon.co.uk/sustainability/

[6] www.canon.co.uk - Printer Ink, Toner & Paper, Genuine Canon Consumables [online] Avaiable at: https://www.canon.co.uk/ink-toner-paper/genuine-consumables/

[7] www.canon.co.uk - Canon Cartridge Recycling Programme, Canon Recycling history [online] Available at: https://www.canon.co.uk/about-us/recycling/

[8] Christie Noe,23 feb 2014- The Manufacture of Printer Ink Cartridges: Materials [online] Available at: https://www.designlife-cycle.com/printer-ink-cartridges#:~:text=This%20casing%20is%20made%20out,Recycling%20Ink%20and%20Toner%20Cartridges).

[9] www.canon.co.uk - Canon Group Directory Europe [online] Available at: https://global.canon/en/corporate/group/europe.html

[10] www.canon.co.uk- Environment, Canon promotes reductions in resource consumption and product-to-product recycling with the aim of contributing to the development of a circular economy. [online] Available at: https://global.canon/en/environment/circulation.html

[11] www.canon.co.uk - Canon recycling inkjet cartridges, FAQs [online] Available at: https://www.canon.co.uk/about-us/recycling/inkjet/